Seller and Inventory Management System

DBMS Project Documentation

1. Introduction

The Seller and Inventory Management System is a database application used to manage sellers, products and stock information. It helps maintain seller-product details, identify available and unavailable products, and generate inventory status reports.

2. Objectives

• Create Seller, Product and Inventory tables.

• Establish relationships between sellers, products and stock.

• Maintain seller and product information.

• Track available and unavailable products.

• Generate inventory status reports using SQL queries.

3. Database Entities

Entity	Purpose
Seller	Stores information about sellers.
Product	Stores information about products supplied by sellers.
Inventory	Stores stock quantity and availability information.

4. Relationship Design

One seller can supply many products. Each product is associated with a seller. Each product has inventory information that records its available stock.

```
Seller (1) ■■■■■■■< Product (M) ■■■■■■■ Inventory (1)
```

seller_id is the primary key in Seller and a foreign key in Product. product_id is the primary key in Product and a foreign key in Inventory.

5. Table Structure

Seller Table	Data Type	Key
seller_id	INT	PRIMARY KEY
seller_name	VARCHAR(100)	
email	VARCHAR(100)	
phone	VARCHAR(15)	

Product Table	Data Type	Key
product_id	INT	PRIMARY KEY
product_name	VARCHAR(100)	
category	VARCHAR(50)	
price	DECIMAL(10,2)	
seller_id	INT	FOREIGN KEY

Inventory Table	Data Type	Key
inventory_id	INT	PRIMARY KEY

product_id	INT	FOREIGN KEY
quantity	INT	
availability	VARCHAR(20)	

6. Database Creation

```
CREATE DATABASE seller_inventory_db;

USE seller_inventory_db;
```

7. Create Seller Table

```
CREATE TABLE Seller (
seller_id INT PRIMARY KEY,
seller_name VARCHAR(100) NOT NULL,
email VARCHAR(100),
phone VARCHAR(15)
);
```

8. Create Product Table

```
CREATE TABLE Product (
product_id INT PRIMARY KEY,
product_name VARCHAR(100) NOT NULL,
category VARCHAR(50),
price DECIMAL(10,2),
seller_id INT,
FOREIGN KEY (seller_id) REFERENCES Seller(seller_id)
);
```

9. Create Inventory Table

```
CREATE TABLE Inventory (
inventory_id INT PRIMARY KEY,
product_id INT,
quantity INT NOT NULL,
availability VARCHAR(20),
FOREIGN KEY (product_id) REFERENCES Product(product_id)
);
```

10. Insert Sample Seller Data

```
INSERT INTO Seller VALUES
(1,'ABC Traders','abc@gmail.com','9876543210'),
(2,'XYZ Stores','xyz@gmail.com','9876501234');
```

11. Insert Sample Product Data

```
INSERT INTO Product VALUES
(101,'Laptop','Electronics',55000,1),
(102,'Keyboard','Electronics',1200,1),
(103,'Office Chair','Furniture',6500,2);
```

12. Insert Inventory Data

```
INSERT INTO Inventory VALUES
(1,101,10,'Available'),
(2,102,0,'Unavailable'),
(3,103,5,'Available');
```

13. Track Available Products

```
SELECT p.product_id, p.product_name, i.quantity
FROM Product p
JOIN Inventory i ON p.product_id = i.product_id
WHERE i.quantity > 0;
```

14. Track Unavailable Products

```
SELECT p.product_id, p.product_name, i.quantity
FROM Product p
JOIN Inventory i ON p.product_id = i.product_id
WHERE i.quantity = 0;
```

15. Seller Product Information

```
SELECT s.seller_name, p.product_name, p.category, p.price
FROM Seller s
JOIN Product p ON s.seller_id = p.seller_id;
```

16. Inventory Status Report

```
SELECT p.product_name, s.seller_name, i.quantity, i.availability
FROM Product p
JOIN Seller s ON p.seller_id = s.seller_id
JOIN Inventory i ON p.product_id = i.product_id;
```

17. Update Stock

```
UPDATE Inventory SET quantity = 15, availability = 'Available' WHERE product_id = 102;
```

18. Delete Product

```
DELETE FROM Product WHERE product_id = 103;
```

Note: A product referenced by Inventory may require deleting/updating the dependent inventory row first, depending on the foreign-key configuration.

19. Conclusion

The Seller and Inventory Management System provides a structured DBMS solution for managing sellers, products and stock. Primary keys uniquely identify records, foreign keys establish relationships, and SQL queries support availability tracking and inventory reporting. The design can be extended with purchase, sales, reorder and supplier modules.